Couldabeens v2: Hardened Models, 1969–2024

couldabeens243 v2 (Phase 2)

1. Replication: v1 models on six more seasons

OLS of `prop_raw ~ year` per era, as in the 2020 report (which had data through 2018 and found the post-era slope positive but insignificant, $p = 0.398$):

Table 1: OLS slope of couldabeen proportion by era, 1969-2024

era	estimate	std.error	p.value
pre	0.0035	0.0010	0.0009
post	0.0002	0.0016	0.9178

2. The right likelihood: binomial GLM with an era interaction

`prop` is `cbns` successes out of `retirees` trials; the proportion enters a quasi-binomial GLM (logit link) and the pre/post slope difference is a fitted interaction rather than two eyeballed fits.

Table 2: Quasi-binomial GLM (dispersion = 1.17). Post-era slope (logit/yr) = 0.0016, SE = 0.0142, z = 0.11

term	estimate	std.error	p.value
(Intercept)	-58.7071	14.9716	0.0003
year	0.0287	0.0075	0.0004
post	53.6294	32.2783	0.1026
year:post	-0.0271	0.0161	0.0975

3. Serial dependence

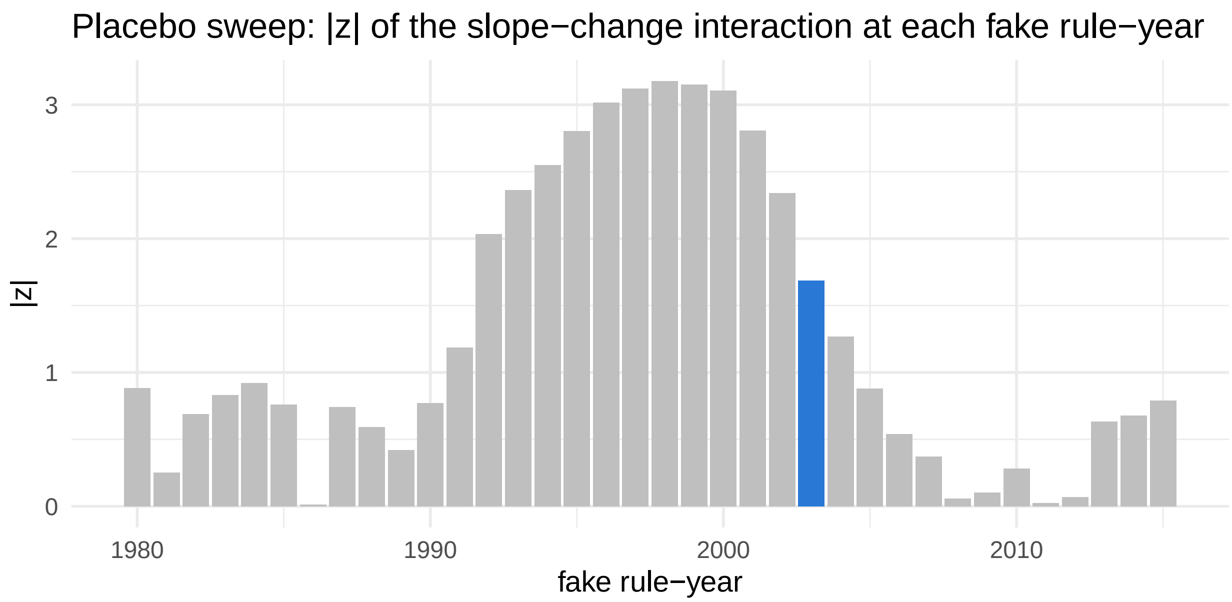
```
## Post-era OLS: Durbin-Watson = 1.69 (p = 0.164).
```

```
## Year slope with Newey-West(3) errors: 0.00016 (SE 0.00145, p = 0.912)
```

4. Earning the 2003 split: changepoint and placebo tests

The 2020 report picked 2003 by eye and confirmed it with a t-test on the same data. Two independent checks: Bai-Perron breakpoint estimation, and a placebo sweep refitting the interaction GLM at every fake rule-year.

```
## Bai-Perron breakpoint estimate(s) (BIC-selected): 1990
```




```
## 2003 ranks 12 of 36 fake rule-years by |interaction z|.
```

5. Labor share, extended and spliced

Post-2003 years with labor-share data (2020 excluded; payroll source splices from the v1 hand-collected series to Spotrac in 2021 – absorbed by a source dummy):

Table 3: Quasi-binomial GLM, post-era, n = 21 years

term	estimate	std.error	p.value
(Intercept)	-1.241	1.404	0.388
labShare	-1.931	3.304	0.566
sourcev1-hand-collected	0.187	0.249	0.461

6. Honest bootstrap

Year-cluster (case) bootstrap of the interaction GLM, B = 2000, seeded:

```
## Post-era logit slope: 0.0016, 95% bootstrap CI [-0.0255, 0.0318]; P(slope > 0) = 0.537
```

7. Sensitivity: classifier and population

Table 4: Post-era slope (logit/yr) across classifiers and populations

series	post_slope	se	z
raw / regulars (headline)	0.002	0.014	0.110
rate / regulars	0.003	0.017	0.191
peak3 / regulars	-0.003	0.010	-0.313
raw / full population	0.018	0.011	1.690

Verdict

With six added seasons and the correct likelihood, the post-2003 trend in the couldabeen share is still not statistically significant (logit slope 0.0016, $z = 0.11$; bootstrap $P(\text{slope}>0) = 0.537$). See the tables above for how this claim moves across classifiers and populations, and the placebo panel for whether 2003 is a distinguished break year rather than an artifact of the split.